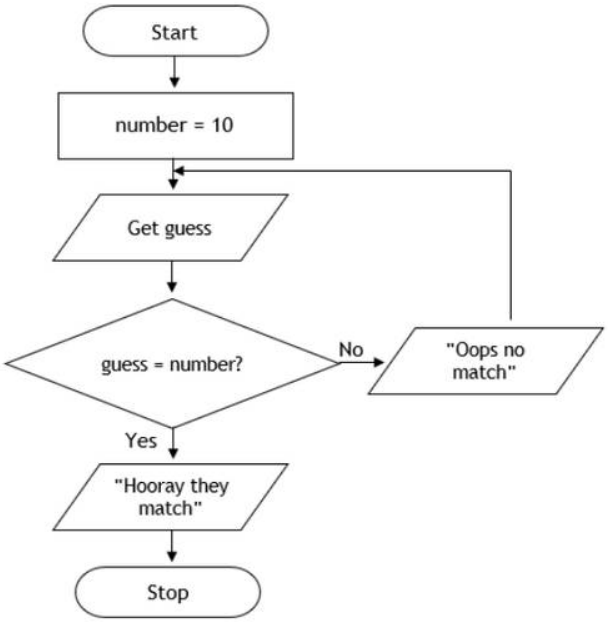
Type MS · Trace tables: run the algorithm by hand

Answer 1 of 13 · Question 4 part c.i · 4 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part c.i in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9 	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Trace tables: run the algorithm by hand

Answer 1 of 13 · Question 4 part c.i · 4 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Trace tables: run the algorithm by hand

Answer 1 of 13 · Question 4 part c.i · 4 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

4(c)(iii)	Award **one** mark for: • Pseudododе that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 2 of 13 · Question 5 part a.i · 3 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

5(a)(i)	One mark for each correct row **Width** / **Length** / **Height** / **Output** 4 / 4 / 2 / Requires large container (1) 2 / 2 / 2 / Too small for cargo container (1) 3 / 8 / 5 / Too large for cargo container (1) Do not penalise spelling	3

Question Number	Answer	Additional Guidance	Mark
5(a)(ii)	Any response that indicates the concept of: Determining which size shipping container is needed for the indicated cargo.		1

Type MS · Trace tables: run the algorithm by hand

Answer 2 of 13 · Question 5 part a.i · 3 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)	SEND states [numStates] TO DISPLAY (1) or SEND state TO DISPLAY (1) SET numStates TO numStates + 1 (1) or LENGTH(states) (1) Concatenation of message and variable uses " and & & numStates & TO DISPLAY (1) or & LENGTH(states) & TO DISPLAY (1) or STRING(numStates) TO DISPLAY (1) Pseudocode is correct (1)	• Accept , or + in place of & • Ignore capitalisation of commands •	4

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

Type MS · Trace tables: run the algorithm by hand

Answer 2 of 13 · Question 5 part a.i · 3 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Showing that index goes up to 4 and does not exceed 4 (1) Showing correspondence between values of index and loadWeight[index] (1)	• There does not have to be an explicit indication of the out-of-bounds / crashing error. • Ignore target and found columns • Allow two rows per loop with matching index and loadWeight values	2

target	found	index	loadWeight[index]
50500	FALSE	0	20000
		1	28000
		2	40000
		3	50000
		4	

target	found	index	loadWeight[index]
50500	FALSE	0	
		1	20000
		2	28000
		3	40000
		4	50000

Question Number	Answer	Additional Guidance	Mark
5(c)(ii)	1. Use of AND operator (in the test of relevant conditions) (1) 2. Use of suitable function to find the length of the array using a correct comparison to index (1) Example: WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO	• Do not penalise pseudocode syntax • Ignore changes to first condition as long as meaning is not lost	2

Type MS · Trace tables: run the algorithm by hand

Answer 2 of 13 · Question 5 part a.i · 3 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 3 of 13 · Question 5 part c.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

5(a)(i)	One mark for each correct row **Width** / **Length** / **Height** / **Output** 4 / 4 / 2 / Requires large container (1) 2 / 2 / 2 / Too small for cargo container (1) 3 / 8 / 5 / Too large for cargo container (1) Do not penalise spelling	3

Question Number	Answer	Additional Guidance	Mark
5(a)(ii)	Any response that indicates the concept of: Determining which size shipping container is needed for the indicated cargo.		1

Type MS · Trace tables: run the algorithm by hand

Answer 3 of 13 · Question 5 part c.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)	SEND states [numStates] TO DISPLAY (1) or SEND state TO DISPLAY (1) SET numStates TO numStates + 1 (1) or LENGTH(states) (1) Concatenation of message and variable uses " and & & numStates & TO DISPLAY (1) or & LENGTH(states) & TO DISPLAY (1) or STRING(numStates) TO DISPLAY (1) Pseudocode is correct (1)	• Accept , or + in place of & • Ignore capitalisation of commands •	4

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

Type MS · Trace tables: run the algorithm by hand

Answer 3 of 13 · Question 5 part c.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Showing that index goes up to 4 and does not exceed 4 (1) Showing correspondence between values of index and loadWeight[index] (1)	• There does not have to be an explicit indication of the out-of-bounds / crashing error. • Ignore target and found columns • Allow two rows per loop with matching index and loadWeight values	2

target	found	index	loadWeight[index]
50500	FALSE	0	20000
		1	28000
		2	40000
		3	50000
		4	

target	found	index	loadWeight[index]
50500	FALSE	0	
		1	20000
		2	28000
		3	40000
		4	50000

Question Number	Answer	Additional Guidance	Mark
5(c)(ii)	1. Use of AND operator (in the test of relevant conditions) (1) 2. Use of suitable function to find the length of the array using a correct comparison to index (1) Example: WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO	• Do not penalise pseudocode syntax • Ignore changes to first condition as long as meaning is not lost	2

Type MS · Trace tables: run the algorithm by hand

Answer 3 of 13 · Question 5 part c.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part b.i in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part b.i in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part b.i in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part b.i in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part b.i in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part b.i in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part b.i in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part b.i in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part b.i in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part b.i in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 4 of 13 · Question 2 part b.i · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part b.i in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part b.iv in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part b.iv in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part b.iv in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part b.iv in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part b.iv in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part b.iv in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part b.iv in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part b.iv in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part b.iv in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part b.iv in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part b.iv in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part b.iv in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 5 of 13 · Question 2 part b.iv · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part b.iv in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 1 of 13 · find part c.v in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 2 of 13 · find part c.v in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 3 of 13 · find part c.v in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 4 of 13 · find part c.v in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 5 of 13 · find part c.v in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 6 of 13 · find part c.v in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 7 of 13 · find part c.v in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 8 of 13 · find part c.v in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 9 of 13 · find part c.v in this mark scheme extract

Q2(c)(v)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 10 of 13 · find part c.v in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 11 of 13 · find part c.v in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 12 of 13 · find part c.v in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 6 of 13 · Question 2 part c.v · 2 marks

2019 JUN · Paper 02 · extract 13 of 13 · find part c.v in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 8 · find part b.i in this mark scheme extract

6(a)	Indicative content provided Other solutions do exist and should be awarded		6

Aspect of Solution	Marks			
	0	1	2	3
Functionality	No awardable content	There are significant errors in logic, leading to an overall solution that is non-functional	There are minor errors in logic, leading to an overall solution that is not completely functional	There are no errors in logic, leading to an overall solution that is fully functional
Accuracy of Notation	No awardable content	Notation follows a broadly unrecognisable convention that is applied inconsistently, although aspects of it are discernible	Notation follows a recognisable convention which is broadly discernible but is applied inconsistently	Notation follows a recognisable convention and is applied consistently throughout

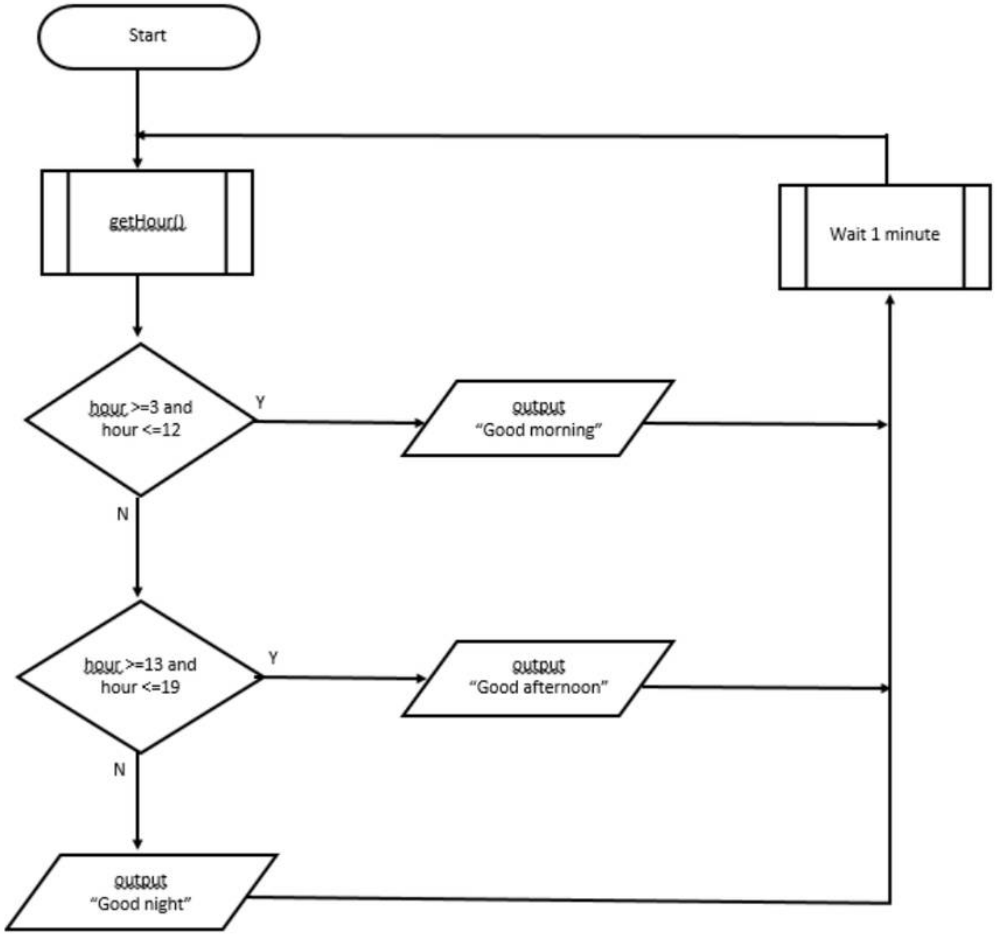
There are a maximum of 3 marks for functionality.
There are a maximum of 3 marks for accuracy of notation.
The marks for functionality and accuracy are awarded independently.

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Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 8 · find part b.i in this mark scheme extract



18

Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	• papaya needs changing (1) • lychee needs ordering (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(ii)	7 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(b)(iii)	Replace 8 with LENGTH(flavours)/LENGTH(volume) (1) or A new line 7: WHILE i < LENGTH(flavours) DO (1) or WHILE i < LENGTH(volume) DO (1)	Ignore case If new line is given award mark if logic is clear	1

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Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 4 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Mark

20

Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 5 of 8 · find part b.i in this mark scheme extract

6(c)	Indicative content: **Compiler** • One line of a compiled language maps to multiple executable instructions • Reads in a whole file and translates it at once • Produces an executable file • Executable file is portable between machines with the same architecture and operating systems • End-user cannot see the programming source code • Does not need an additional environment/software to run the code **Interpreter** • One line of a compiled language maps to multiple executable instructions • Reads, translates, and executes one line at a time • A special environment is needed to be installed on the user's machine to run the code; this is machine specific • Source code is portable to any machine with an interpreter which can run on it • End-user can see the programming source code **Similarities** • One line of a compiled language maps to multiple executable instructions/one to many relationship to machine code • Both are used to interpret high level programming languages • Both produce machine code from human-readable programming code **Differences** • Compiler reads in a whole file and translates it at once. Interpreter reads, translates and executes one line at a time • Compiler produces an executable file, interpreter does not • Executable files produced by a compiler will only execute on machines with the same architecture and operating systems. Source code produced by an interpreter is portable to any machine with an interpreter which can run on it • Compiled code does not require any additional software	6

Type MS · Trace tables: run the algorithm by hand

Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 6 of 8 · find part b.i in this mark scheme extract

	to execute the code. An interpreter needs a special environment to be installed on the user's machine to run the code; this is machine specific	

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Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 7 of 8 · find part b.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 7 of 13 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 8 of 8 · find part b.i in this mark scheme extract

Type MS · Trace tables: run the algorithm by hand

Answer 8 of 13 · Question 5 part a.i · 5 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td></td><td></td><td>10</td><td>3</td><td></td><td></td><td></td><td>10</td></tr>
<tr><td>2</td><td></td><td></td><td>6</td><td></td><td>9</td><td></td><td>6</td><td></td></tr>
</table>

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
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<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td>4</td><td>2</td><td>10</td><td>3</td><td>6</td><td>1</td><td>9</td><td>10</td></tr>
<tr><td>2</td><td>4</td><td>2</td><td>6</td><td>3</td><td>9</td><td>1</td><td>6</td><td>10</td></tr>
</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Trace tables: run the algorithm by hand

Answer 8 of 13 · Question 5 part a.i · 5 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Trace tables: run the algorithm by hand

Answer 8 of 13 · Question 5 part a.i · 5 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 8 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 8 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: <table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table>		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 3 of 8 · find part d.i in this mark scheme extract

Question	IMP	Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 4 of 8 · find part d.i in this mark scheme extract

		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 5 of 8 · find part d.i in this mark scheme extract

		• plaintextLetter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 6 of 8 · find part d.i in this mark scheme extract

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 7 of 8 · find part d.i in this mark scheme extract

Question	MP	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
15
16 #       Add your code to get the plaintext and convert it to lowercase
17 plaintext = input("Enter the plaintext using lowercase letters: ")
18 plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
39 # Add your code to write the ciphertext to a text file
40 cipherFile = open("Cipher.txt", "w")
41 cipherFile.write(ciphertext)
42 cipherFile.close()
```

```
40
41 #       Add your code to display the ciphertext
42 print("The ciphertext message is:",ciphertext)
43
```

Type MS · Trace tables: run the algorithm by hand

Answer 9 of 13 · Question 3 part d.i · 1 mark

2021 JUN · Paper 02 · extract 8 of 8 · find part d.i in this mark scheme extract

Type MS · Trace tables: run the algorithm by hand

Answer 10 of 13 · Question 4 part b.i · 5 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

<table>
<tr>
<td>4(a)</td>
<td>Award one mark from:

• a step-by-step description of a process that completes a task (1)
• a set of instructions that describes how to get something done (1)</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>4(b)(i)</td>
<td>Award one mark each to a maximum of five marks for:

• two correct decisions in last two lines (1)
• two correct outputs (1)
• correct increment at Count=Count+1 (1)
• correct change for Number3=Number2+Number1 (1)
• correct change for Number1=Number2 AND Number2=Number3 (1)

<table>
<tr><th>Number 1</th><th>Number 2</th><th>Number 3</th><th>Count</th><th>Output</th><th>Count = 2?</th></tr>
<tr><td>1</td><td>1</td><td>-</td><td>0</td><td>1 1</td><td>False</td></tr>
<tr><td>1</td><td>2</td><td>2</td><td>1</td><td>2</td><td>False</td></tr>
<tr><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td>True</td></tr>
</table>
</td>
<td>Ignore spelling errors

Allow true false, Y N etc. for decisions</td>
<td>5</td>
</tr>
</table>

Type MS · Trace tables: run the algorithm by hand

Answer 10 of 13 · Question 4 part b.i · 5 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

4(b)(ii)	Award **one** mark each for: • helps visualise how the algorithm works (1) • helps detect (logic) errors (1)		1

Question Number	Answer	Additional Guidance	Mark
4(c)	Award **one** mark each to a maximum of **five** marks for: • SET count TO 0 (1) • IF statement checks for a match to 1 vowel (1) • IF statement checks for a match to all vowels (1) • count incremented correctly (1) • display count (1) • concatenates suitable message with the count (1) SET word TO "elephant" SET count TO 0 FOR EACH letter FROM word DO IF letter = 'a' OR letter = 'e' OR letter = 'i' OR letter = 'o' OR letter = 'u' THEN SET count TO count + 1 END IF END FOREACH SEND 'The number of vowels is ' & count TO DISPLAY		5
		Total	**12**

Type MS · Trace tables: run the algorithm by hand

Answer 10 of 13 · Question 4 part b.i · 5 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

3(a)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • Each column represents a variable/output (1) • Rows shows how the values of variables change (as the code is executed) (1) • Values are filled in manually/by hand (1)	Line numbers are for guidance – mark answer as a whole	**2**

Question	mp	Answer	Additional Guidance	Mark
3(b)	B1 B2 B3	Award **one** mark for each correct cell: **num1** | **num2** | **Output** 88 | 18 | State 2 17 | 18 | State 4 12 | 19 | State 1	ignore quotes e.g. "State 2"	**3**

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
3(c)	Award **one** mark for each of:		C4 should exit loop Allow C5 if a loop is used to keep asking for a non-negative exponent C9 – ignore error message in C6; must have base, exponent, and answer in output for this mark	**10**
	C1	At least one variable with a suitable variable name		
	C2	At least one variable initialised to an appropriate value		
	C3	While loop used to continue running code		
	C4	Relational test to identify terminating condition for loop		
	C5	Selection used to check for negative exponent		
	C6	Appropriate error message if exponent is negative		
	C7	Calculation of answer using any method		
	C8	A loop used to calculate answer as indicated in flowchart		
	C9	Prompts and output messages are fit for purpose		
	C10	Executing and producing the correct output for anticipated inputs		

Suggested test data and output		
Base	**Exponent**	**Output**
0		ends program
2	0	2 to power of 0 is 1
3	4	3 to power of 4 is 81

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

C#

```
// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

// Display the first prompt and get the base number
Console.Write("Enter a base number (0 to exit)): ");
baseNum = Convert.ToInt32(Console.ReadLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    Console.Write("Enter an exponent number: ");
    exponent = Convert.ToInt32(Console.ReadLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        Console.Write("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i = 0; i < exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = baseNum.ToString() + " to the power of " +
                    exponent.ToString() + " is " +
                    answer.ToString();
        Console.Write(outString);
    }
    // Display the first prompt again and get the base number
    Console.Write("\nEnter a base number (0 to exit)): ");
    baseNum = Convert.ToInt32(Console.ReadLine());
}
```

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract

Java

```
// --------------------------------------------------
// Write your code below this line

// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

Scanner myScanner = new Scanner (System.in);

// Display the first prompt and get the base number
System.out.print("Enter a base number (0 to exit)): ");
baseNum = Integer.parseInt (myScanner.nextLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    System.out.print("Enter an exponent number: ");
    exponent = Integer.parseInt (myScanner.nextLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        System.out.println ("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i=0; i<exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = Integer.toString(baseNum) + " to the power of " +
                    Integer.toString(exponent) + " is " +
                    Integer.toString(answer);
        System.out.println(outString);
    }

    // Display the first prompt again and get the base number
    System.out.print("Enter a base number (0 to exit)): ");
    baseNum = Integer.parseInt(myScanner.nextLine());
}
```

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

Python

```
# Initialise variables
base = 0
exponent = 0
answer = 1

# Get the first base number
base = int (input ("Enter a base number (0 to exit): "))

# Check if the user wants to exit
while (base != 0):

    # Get the exponent number
    exponent = int (input ("Enter an exponent number: "))

    # Check if the exponent is negative
    if (exponent < 0):
        print ("Invalid exponent entered")
    else:
        # Reset the answer
        answer = 1

        # Calculate the answer using exponentiation
        for count in range (0, exponent):
            answer = answer * base

        # Create a string for the output sentence
        print (str (base) + " to the power of " +
               str (exponent) + " is " +
               str (answer))

    # Get another base number and lp
    base = int (input ("Enter a base number (0 to exit): "))
```

Type MS · Trace tables: run the algorithm by hand

Answer 11 of 13 · Question 3 part b · 3 marks

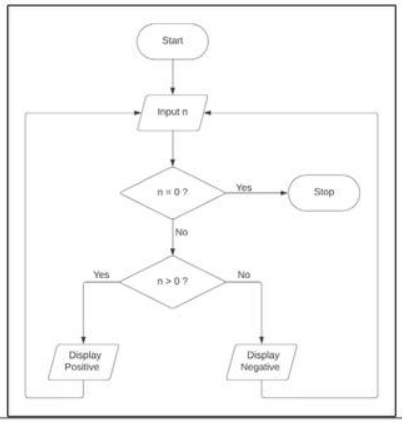
2023 JUN · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Trace tables: run the algorithm by hand

Answer 12 of 13 · Question 6 part b.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	mark
6(a)	Award **one** mark for each of: • Start and Input n identified (accept number or num) (1) • Decision box test for n = 0/n<>0 with yes & no/true & false labels that correctly lead to End/Stop (1) • Decision box test for n > 0 or n<0 with yes & no/true & false labels (1) • Correct messages output including leads from correct yes & no/true & false (1) • Flowchart is fully connected and functionally correct (1) • Correct flowchart symbols used throughout for terminators, input/output and decision boxes (1) Example:  	Accept == for MP2 For MP2 and MP3 decision boxes must have yes/no or true/false labels. For MP5 there must be arrows on connections	6

Type MS · Trace tables: run the algorithm by hand

Answer 12 of 13 · Question 6 part b.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(i)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>7</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td></tr><tr><td></td><td></td><td>4</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td></tr><tr><td></td><td></td><td>6</td><td></td></tr><tr><td></td><td>TRUE</td><td>7</td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 7 FALSE 2</td><td>2</td></tr>
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</table>

Type MS · Trace tables: run the algorithm by hand

Answer 12 of 13 · Question 6 part b.i · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
6(b)(iii)	Award **one** mark for: • To determine if a number is a prime number/none of the test numbers divide exactly into it (1)	Ignore reference to display the number	1
Question	**Answer**	**Additional Guidance**	**Mark**
6(b)(iv)	Award up to **two** marks for a linked explanation such as: • It makes the algorithm more efficient (1) because the loop will terminate as soon as a test number divides exactly into the number entered (1) • A count-controlled loop could make more comparisons/be less efficient (1) because it would not terminate as soon as the test number divides exactly into the number entered (1) • A count-controlled loop executes a fixed number of times/even after condition is met (1) post-condition loop stops when condition is met (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 6	**13**

Type MS · Trace tables: run the algorithm by hand

Answer 12 of 13 · Question 6 part b.i · 2 marks

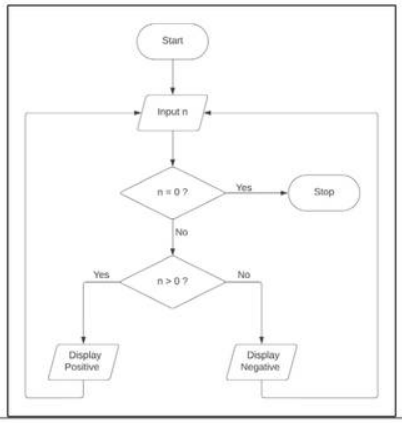
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Answer 13 of 13 · Question 6 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	mark
6(a)	Award **one** mark for each of: • Start and Input n identified (accept number or num) (1) • Decision box test for n = 0/n<>0 with yes & no/true & false labels that correctly lead to End/Stop (1) • Decision box test for n > 0 or n<0 with yes & no/true & false labels (1) • Correct messages output including leads from correct yes & no/true & false (1) • Flowchart is fully connected and functionally correct (1) • Correct flowchart symbols used throughout for terminators, input/output and decision boxes (1) Example:  	Accept == for MP2 For MP2 and MP3 decision boxes must have yes/no or true/false labels. For MP5 there must be arrows on connections	6

Type MS · Trace tables: run the algorithm by hand

Answer 13 of 13 · Question 6 part b.ii · 2 marks

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<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(i)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>7</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td></tr><tr><td></td><td></td><td>4</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td></tr><tr><td></td><td></td><td>6</td><td></td></tr><tr><td></td><td>TRUE</td><td>7</td><td></td></tr></table></td><td>Accept for first pair
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<tr><td>6(b)(ii)</td><td>Award one mark for each pair of columns:
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Type MS · Trace tables: run the algorithm by hand

Answer 13 of 13 · Question 6 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
6(b)(iii)	Award **one** mark for: • To determine if a number is a prime number/none of the test numbers divide exactly into it (1)	Ignore reference to display the number	1
Question	**Answer**	**Additional Guidance**	**Mark**
6(b)(iv)	Award up to **two** marks for a linked explanation such as: • It makes the algorithm more efficient (1) because the loop will terminate as soon as a test number divides exactly into the number entered (1) • A count-controlled loop could make more comparisons/be less efficient (1) because it would not terminate as soon as the test number divides exactly into the number entered (1) • A count-controlled loop executes a fixed number of times/even after condition is met (1) post-condition loop stops when condition is met (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 6	**13**

Type MS · Trace tables: run the algorithm by hand

Answer 13 of 13 · Question 6 part b.ii · 2 marks

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